

Lee, Doehyeon

Tel: +82 10 4689 0106 | E-mail: anthony16@snu.ac.kr | Address: Seoul National University | [Homepage](#)

EDUCATION

MS, Industrial Engineering, Seoul National University (GPA: 4.06/4.30) Mar. 2025 – Feb. 2027 Expected

- Supervisor: Prof. [Jonghun Park](#) ([Information Management Lab](#))

BS, Industrial Engineering, Seoul National University (GPA: 3.76/4.30) Mar. 2019 – Feb. 2025

PUBLICATIONS

[ICML 2025 Poster] ELITE: Enhanced Language-Image Toxicity Evaluation for Safety ([Arxiv link](#))

- **(Co-First Author)** Proposed the **ELITE benchmark** to address insufficient harmfulness coverage and ambiguity in existing safety evaluation benchmarks for Vision-Language Models (VLMs).
- Developed the rubric-based evaluator that explicitly integrates a Toxicity Score to achieve superior alignment with human evaluations in assessing multimodal harmfulness.
- Contributed to the development of safer VLMs by systematically evaluating and mitigating safety risks.

[ICLR 2026 Poster] Jailbreaking on Text-to-Video Models via Scene Splitting Strategy ([Arxiv link](#))

- **(Co-author)** Contributed to the development of **SceneSplit**, a novel **black-box jailbreak method** that splits narratives to target major Text-to-Video models (Hailuo, Sora, etc.), significantly outperforming existing baselines.
- Developed a mechanism to manipulate the generative output space by fragmenting harmful narratives into sequential benign scenes to bypass safety filters, integrating strategies for enhanced execution.

RESEARCH PROJECTS

Nanyang Technological University Global Connect Fellowship Feb. 2026 – Mar. 2026

- Conducted research at the [Data Privacy and Security Lab](#) (Supervisor: Prof. [Wei Dong](#)), **NTU Singapore**, investigating perspective bias in LLMs across real-world tasks.
- Applied Sparse Autoencoders (SAEs) to decompose internal feature representations and identify latent directions that drive biased outputs.

AI Toolkit for Composer: Improving Text-to-Symbolic Music Generation Models via Synthetic Data

Lead Researcher Sep. 2025 – Aug. 2026 (Ongoing)

- Funded by the National Research Foundation of Korea (NRF) M.S. Research Scholarship (No. RS-2025-25424095).
- Developing a generative AI framework for text-to-symbolic music generation using synthetic datasets, with parallel investigation into AI-generated music detection to evaluate the authenticity of generated outputs.

WORK EXPERIENCE

AIM Intelligence ([link](#)) *Research Intern* Oct. 2024 – Feb. 2025

- Designed and built a VLM Safety Benchmark with improved data categorization and quality (ICML 2025 Poster).
- Developed novel jailbreak method for Text-to-Video models with enhanced attack success rate (ICLR 2026 Poster).
- Executed red teaming and jailbreak testing for an LLM-based care-call service, identifying key vulnerabilities.
- Continuously monitored and analyzed AI security trends, focusing on the latest models and emerging risks.

NAVER Cloud ([link](#)) *Intern, Hyperscale AI planning, Eco biz Dev* Jul. 2024 – Sep. 2024

- Analyzed market trends in GenAI services and AI Agents, leveraging collected data to propose strategic directions.
- Formulated partnership strategies and developed business development roadmaps for key service sectors, contributing to ecosystem expansion initiatives.

Music & Audio Research Group, Seoul National University *Research Intern* Jul. 2023 – Aug. 2023

- Supervisor: Prof. [Kyogu Lee](#) ([Music & Audio Research Group](#))
- Developed and optimized a CNN-based chorus detection model for automated musical structure analysis.

TEACHING EXPERIENCE

Industrial Application of Machine Learning *Teaching Assistant* Sep. 2025 – Dec. 2025

- Graduate-level core course, Department of Industrial Engineering, Seoul National University

Basic Computing: First Adventures in Computing *Tutor* Mar. 2025 – Dec. 2025

- Introductory undergraduate course, SNU College (College of Liberal Studies), Seoul National University